

14	B	Total Energy = $\frac{1}{2} m \lambda_0^2 \omega^2 = \frac{1}{2} (0.00)(0.15) (2\pi/3.0)^2 = 1.1 \times 10^{-3} \text{ J}$
15	A	Further damping causes amplitude to decrease at all frequencies
16	D	$I = \frac{P}{4\pi r^2}$ At 3.0 m, $P = 0.18 \times 4\pi \times 9 = 6.48\pi$ Power is now tripled, new $P = 19.44\pi$ At 4.5 m, $I = \frac{P}{4\pi r^2} = \frac{19.44\pi}{4\pi 4.5^2} = 0.24 \text{ W m}^{-2}$
17	C	Stretched string has a node at the two ends with a antinode at the centre (fundamental mode). The wavelength of the sound is equivalent to $2L$. Only Option C gives us the